G M Tasnim Alam

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Summary

Ph.D. candidate (Computer Science) specializing in cybersecurity research, focusing on protocol security, fuzzing, and vulnerability detection. Industry experience at Apple as an Intern and Agoda (a Booking Holdings company) as a Senior Software Engineer shipping production software and automation. Strong in Python/C++/Swift, debugging, distributed pipelines (Kafka/ELK/Docker), and CI/CD. Seeking Software Engineering or Research Intern roles in Systems, Backend, Security, or Infrastructure.

Education

State University of New York at Stony Brook — *New York, United States*
Ph.D. Candidate in Computer Science — *Fall 2023 – Present*

- Coursework: Systems Security, Network Security, Algorithms, Big Data, Data Science, Visualization

Skills

Languages:	Python, C++, Swift, Java
Systems & DevOps:	Microservice, Git, iOS, Docker, GitHub Actions, CI/CD
Data/Infra & Observability:	Kafka, Elasticsearch, Logstash, REST APIs, osquery, eBPF

Experience

Apple — *California, USA*
Software Engineering Intern — *May 2025 – Aug 2025*

- Built an accessibility feature improving usability for customers with limited mobility; delivered a production-ready implementation spanning design, development, and testing.

Agoda (Booking Holdings) — *Bangkok, Thailand*
Senior Software Engineer — *Jan 2022 – June 2023*

- Designed and delivered a shopping-cart and multi-product checkout module using modular architecture patterns; improved checkout conversion by 15%.
- Built booking workflows and integrated with APIs and analytics, improving reliability of the purchase funnel and reducing user friction.
- Migrated analytics components from C# to a native implementation, improving data correctness and processing efficiency.

Publication

ACM WiSec 2025: *Performance and Consistency Trade-off of the eSIM M2M Remote Provisioning Protocol.*
https://dl.acm.org/doi/abs/10.1145/3734477.3734712

CVEs / Vulnerability Disclosures

CVE-2026-30075, CVE-2026-30077, CVE-2026-30078, CVE-2026-30079, CVE-2026-30080, CVE-2026-5360, CVE-2026-4531, CVE-2026-5661

Projects

Black-Box Resiliency Analyzer for 5G Core

- Built an adaptive fuzzing and resiliency testing tool for 5G Core networks that mutates NGAP messages, exercises stateful scenarios, and flags/classifies bugs based on system behavior and responses.

CI/CD Pipeline Security Analysis

- Built a DevSecOps monitoring pipeline for GitHub Actions by collecting host/workflow telemetry with osquery and eBPF, streaming events via Kafka, and indexing with Logstash/Elasticsearch to support detection, auditing, and incident triage.

Malicious Repository Detection (Big Data)

- Implemented scalable data pipelines and detection heuristics to identify potentially malicious repositories using big-data techniques.

NYC Motor Vehicle Collision Analysis

- Developed a data analytics and visualization pipeline for NYC vehicle collision datasets using Python (data cleaning/feature engineering) and D3.js (interactive dashboards) to surface trends, hotspots, and risk patterns.

Constraint-Based Text Generation (Gatsby Without "e")

- Designed an NLP generation pipeline using LLMs to produce constrained text outputs (character-level restrictions), including data preprocessing, prompt/constraint enforcement, and automated evaluation of generated results.

Teaching Experience

Teaching Assistant *Stony Brook University*

CSE 541: Logic in Computer Science | CSE 381: Game Programming | CSE 416: Software Engineering

Academic Service

Sub-reviewer, IEEE Symposium on Security and Privacy (S&P), 2025

Awards & Fellowships

Cyber Powder Fellow, University of Utah, 2025